

Elia Gatti

Software Engineer (HPC & GPU) · M.S. Computer Science, University of Trento

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Available Mar 2027

Work Experience

Bitmovin, Software Engineer Intern

June 2026 to Sept 2026

- Shipped KAIROS **end-to-end**, an AI video highlight detection and segmentation service (VOD plus live): a **public API with MCP** and Go/Python workers on Terraform-managed GCP. [demo]
- Cut transcription latency by **20%** by tuning worker parallelism and slimming workers, guided by live playback measurements.
- Reduced failed runs and manual operations by building **job orchestration** with startup fixes and **live observability**.
- Backed the API and workers with a **Supabase/PostgreSQL schema**, queries, and **RLS policies**.

Dedagroup, Software Engineer Intern

May 2024 to Sept 2024

- Halved VM resource requirements by migrating the 'TEN' treasury application from legacy Windows to Linux servers.
- Delivered full-stack features with an HTMX frontend on a Java/Spring backend while maintaining the existing codebase.

Education

M.S. in Computer Science, University of Trento

Expected Mar 2027

Thesis (in progress): GPU kernel optimization. GPU programming and high-performance computing.

B.S. in Computer Science, University of Trento

2024

Bachelor's thesis: MPEG-DASH performance analysis under simulated (Mininet) and real (AWS) networks.

Projects

Sparse Matrix-Vector Multiplication on GPU [code]

- Developed a **hybrid adaptive CUDA kernel** for NVIDIA A30, plus a V2 with sub-warp lane classes and huge-row splitting (deterministic finalize, no atomics).
- Verified results against a double-precision CPU reference while profiling execution time and GFLOPS (memory coalescing, occupancy, L2-cache control).

Parallel Minimum Spanning Tree (MPI + OpenMP) [code]

- Implemented parallel Boruvka/Kruskal MST with **MPI + OpenMP** using a flat 1D layout, mmap I/O, and bulk collectives; analyzed speedup and scalability on a CPU cluster (report in repo).

Distributed Key-Value Storage System [code]

- Built a fault-tolerant store with **Akka actors** (Java 21) using consistent hashing and quorum reads/writes, surviving concurrent access and node failures without client disruption.

Achievements

CTM Challenge, Trento 2025 — 1st place [code]

- Engineered SVD-based watermark embedding within DWT blocks, robust to compression, noise and transforms; validated with an automated attack suite using ROC analysis.

EuroTech HK Hackathon, Munich 2026 — Guardian: privacy-first eldercare monitor [code]

- Built zero-config elderly monitoring with mmWave radar plus an on-device Gemma 4 LLM (no cameras, wearables, or cloud) with behavioral drift tracking and real-time crisis alerting.

Skills

Languages: C, C++, CUDA, Java, Python, Go

Tools & Frameworks: MPI, OpenMP, Slurm, Docker, Git, Linux, PostgreSQL, FastAPI

Spoken languages: Italian (native), English (fluent)